Language Modeling

Approximating Shakespeare

1 gram	–To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have –Hill he late speaks; or! a more to leg less first you enter
2 gram	–Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow. –What means, sir. I confess she? then all sorts, he is trim, captain.
3 gram	–Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, 'tis done. –This shall forbid it should be branded, if renown made it empty.
4 gram	–King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv'd in; –It cannot be but so.

Approximating Wall Street Journal

1 gram	Months the my and issue of year foreign new exchange's september were recession exchange new endorsed a acquire to six executives
2 gram	Last December through the way to preserve the Hudson corporation N. B. E. C. Taylor would seem to complete the major central planners one point five percent of U. S. E. has already old M. X. corporation of living on information such as more frequently fishing to keep her
3 gram	They also point to ninety nine point six billion dollars from two hundred four oh six three percent of the rates of interest stores as Mexico and Brazil on market conditions

Problems with MLE Models

- MLE models maximizes the probability on the observed training data and do not waste any probability mass on unobserved events

- If no probability mass is assigned to unseen events, then all sentences with unseen events all get a probability of 0

- Power-law frequency distribution: it is very likely to encounter unseen words in unseen text, even more so unseen n-grams

- Need method to account for unseen events!

Zipf’s law

- Zipf's law is an empirical law, that models the frequency distribution of words in languages. And by languages, this is not just restricted to English corpora.

- According to Zipf's law, the frequency of a given word is dependent on the inverse of it's rank

$$f(r, \alpha) \propto 1/r^{\alpha}$$

Here,
$\alpha = 1$, r = rank of a word, $f(r, \alpha)$ = frequency in the corpus

- The word with the second highest rank will have the frequency value ½ of the first word, the word with the third highest rank will have the frequency value ⅓ of the first word, and so on...

Most frequent words

- Noun
  - Time, Person, Year, Way, Day
- Verb
  - be, have, do, say, get
- Adjectives
  - Good, new, first, last, long
- Overall
  - The, be, to, of, and

Zeros : Issues???

Training set:

I am Sam
Sam I am
I do not like green eggs and ham

Test set

You hate vegetables

Training set:

… denied the allegations.
… denied the reports
… denied the claims
… denied the request

P("offer" | denied the) = 0

Test set

… denied the offer
… denied the loan

Zero probability Unigrams

- Approach 1: Turn the problem back into a closed vocabulary one by choosing a fixed vocabulary in advance:
  1. Choose a vocabulary (word list) that is fixed in advance.
  2. Convert in the training set any word that is not in this set (any OOV word) to the unknown word token <UNK> in a text normalization step.
  3. Estimate the probabilities for <UNK> from its counts just like any other regular word in the training set.

- Approach 2: Replacing words in the training data by <UNK> based on their frequency.

Zero probability Bigrams

- Bigrams with zero probability mean that we will assign 0 probability to the test set!

- And hence we cannot compute perplexity (can’t divide by 0)!

Smoothing

- Smoothing is a way to deal with unobserved n-grams

- Works by taking a little bit of the probability mass from higher counts and shift it to zero counts

- A closed vocabulary, i.e. no unseen words, but unseen n-grams only

Add-one estimation / Laplace smoothing

- Idea: We add 1 to all possible frequency counts

For vocabulary size V:

unigram $$P_{Lap}(w) = \frac{C(w)+1}{N+V}$$

bigram $$P_{Lap}(w_i \mid w_{i-1}) = \frac{C(w_{i-1},w_i)+1}{C(w_{i-1})+V}$$

Berkeley Restaurant Corpus: Laplace smoothed bigram counts

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

	i	want	to	eat	chinese	food	lunch	spend
i	6	828	1	10	1	1	1	3
want	3	1	609	2	7	7	6	2
to	3	1	5	687	3	1	7	212
eat	1	1	3	1	17	3	43	1
chinese	2	1	1	1	1	83	2	1
food	16	1	16	1	2	5	1	1
lunch	3	1	1	1	1	2	1	1
spend	2	1	2	1	1	1	1	1

Laplace-smoothed Bigram Probabilities

$$P^*(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)+1}{C(w_{n-1})+V}$$

	i	want	to	eat	chinese	food	lunch	spend
i	0.0015	0.21	0.00025	0.0025	0.00025	0.00025	0.00025	0.00075
want	0.0013	0.00042	0.26	0.00084	0.0029	0.0029	0.0025	0.00084
to	0.00078	0.00026	0.0013	0.18	0.00078	0.00026	0.0018	0.055
eat	0.00046	0.00046	0.0014	0.00046	0.0078	0.0014	0.02	0.00046
chinese	0.0012	0.00062	0.00062	0.00062	0.00062	0.052	0.0012	0.00062
food	0.0063	0.00039	0.0063	0.00039	0.00079	0.002	0.00039	0.00039
lunch	0.0017	0.00056	0.00056	0.00056	0.00056	0.0011	0.00056	0.00056
spend	0.0012	0.00058	0.0012	0.00058	0.00058	0.00058	0.00058	0.00058

Bigram Probabilities

	i	want	to	eat	chinese	food	lunch	spend
i	0.002	0.33	0	0.0036	0	0	0	0.00079
want	0.0022	0	0.66	0.0011	0.0065	0.0065	0.0054	0.0011
to	0.00083	0	0.0017	0.28	0.00083	0	0.0025	0.087
eat	0	0	0.0027	0	0.021	0.0027	0.056	0
chinese	0.0063	0	0	0	0	0.52	0.0063	0
food	0.014	0	0.014	0	0.00092	0.0037	0	0
lunch	0.0059	0	0	0	0	0.0029	0	0
spend	0.0036	0	0.0036	0	0	0	0	0

Before Smoothing

	i	want	to	eat	chinese	food	lunch	spend
i	0.0015	0.21	0.00025	0.0025	0.00025	0.00025	0.00025	0.00075
want	0.0013	0.00042	0.26	0.00084	0.0029	0.0029	0.0025	0.00084
to	0.00078	0.00026	0.0013	0.18	0.00078	0.00026	0.0018	0.055
eat	0.00046	0.00046	0.0014	0.00046	0.0078	0.0014	0.02	0.00046
chinese	0.0012	0.00062	0.00062	0.00062	0.00062	0.052	0.0012	0.00062
food	0.0063	0.00039	0.0063	0.00039	0.00079	0.002	0.00039	0.00039
lunch	0.0017	0.00056	0.00056	0.00056	0.00056	0.0011	0.00056	0.00056
spend	0.0012	0.00058	0.0012	0.00058	0.00058	0.00058	0.00058	0.00058

After Smoothing

Add-k smoothing

$$P^*_{\text{Add-k}}(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)+k}{C(w_{n-1})+kV}$$

K is a fraction —— 0.5, 0.05, 0.01,……

Backoff

- If we are trying to compute trigram probability, but we have no examples of a particular trigram
  - we can instead estimate its probability by using the bigram probability
  - similarly, if we don’t have counts to compute bigram probability we can look to the unigram.

Interpolation

- Linear interpolation (mixture model): Combine probabilities using a linear combination

$$P_{li}(w_n \mid w_{n-2}w_{n-1}) = \lambda_1 P_1(w_n) + \lambda_2 P_2(w_n \mid w_{n-1}) + \lambda_3 P_3(w_n \mid w_{n-2}w_{n-1}) \text{ where } 0 \le \lambda_i \le 1 \text{ and } \sum_i \lambda_i = 1$$

In a nutshell

- n-gram language models are a simple way to represent local regularities of language
  - they can be trained from raw text
  - they do not account for long-range dependencies
  - they do not account for grammatical phenomena

- Applications of language models:
  - similarity of document collections
  - language generation post-processing (MT)
  - Information Retrieval

  ………..

Exercise

Exercise 1 *Consider the following toy example*

Training data:

```
<s> I am Sam </s>
<s> Sam I am </s>
<s> Sam I like </s>
<s> Sam I do like </s>
<s> do I like Sam </s>
```

Assume that we use a bigram language model based on the above training data.

1. *What is the most probable next word predicted by the model for the following word sequences?*

   (1) `<s>` *Sam ...*

   (2) `<s>` *Sam I do ...*

   (3) `<s>` *Sam I am Sam ...*

   (4) `<s>` *do I like ...*

2. *Which of the following sentences is better, i.e., gets a higher probability with this model?*

   (5) `<s>` *Sam I do I like* `</s>`

   (6) `<s>` *Sam I am* `</s>`

   (7) `<s>` *I do like Sam I am* `</s>`

Exercise - 1 - Solution Sketch

Bigram probabilities:

$P(\texttt{Sam}|\texttt{<s>}) = \frac{3}{5}$ $P(\texttt{I}|\texttt{<s>}) = \frac{1}{5}$
$P(\texttt{I}|\texttt{Sam}) = \frac{3}{5}$ $P(\texttt{</s>}|\texttt{Sam}) = \frac{2}{5}$
$P(\texttt{Sam}|\texttt{am}) = \frac{1}{2}$ $P(\texttt{</s>}|\texttt{am}) = \frac{1}{2}$
$P(\texttt{am}|\texttt{I}) = \frac{2}{5}$ $P(\texttt{like}|\texttt{I}) = \frac{2}{5}$ $P(\texttt{do}|\texttt{I}) = \frac{1}{5}$
$P(\texttt{Sam}|\texttt{like}) = \frac{1}{3}$ $P(\texttt{</s>}|\texttt{like}) = \frac{2}{3}$
$P(\texttt{like}|\texttt{do}) = \frac{1}{2}$ $P(\texttt{I}|\texttt{do}) = \frac{1}{2}$

1. (1) and (3): "I".

   (2): "I" and "like" are equally probable.

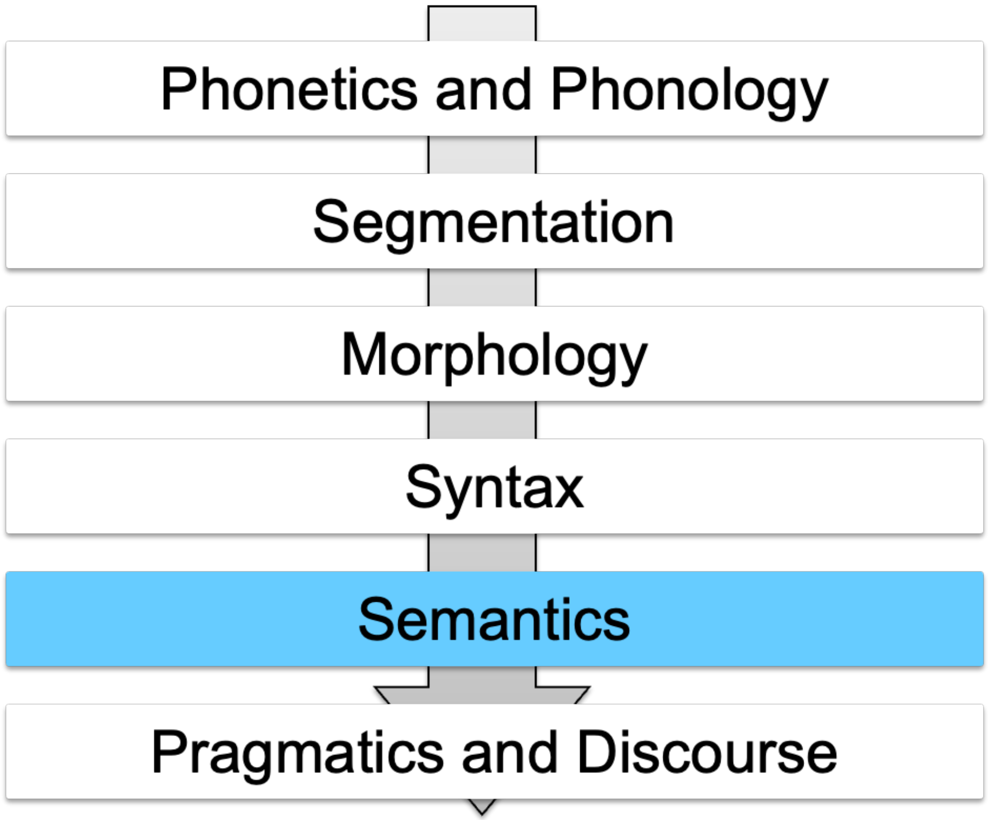
   (4): </s>

2. Probabilities:

   (5): $\frac{3}{5} \cdot \frac{3}{5} \cdot \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{2}{5} \cdot \frac{2}{3}$
   (6): $\frac{3}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{2}$
   (7): $\frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{2}$
   (6) is the most probable sentence according to our language model.

Exercise 2 *Consider again the same training data and the same bigram model. Compute the perplexity of*

I do like Sam


Phonetics and Phonology
Segmentation
Morphology
Syntax
Semantics
Pragmatics and Discourse

What is Semantics?

Study of the meaning of **words, phrases, sentences, or documents**

What is the meaning of a word?

Stone

Unicorn

Greed

What is the meaning of a word?

Stone

Unicorn

Greed

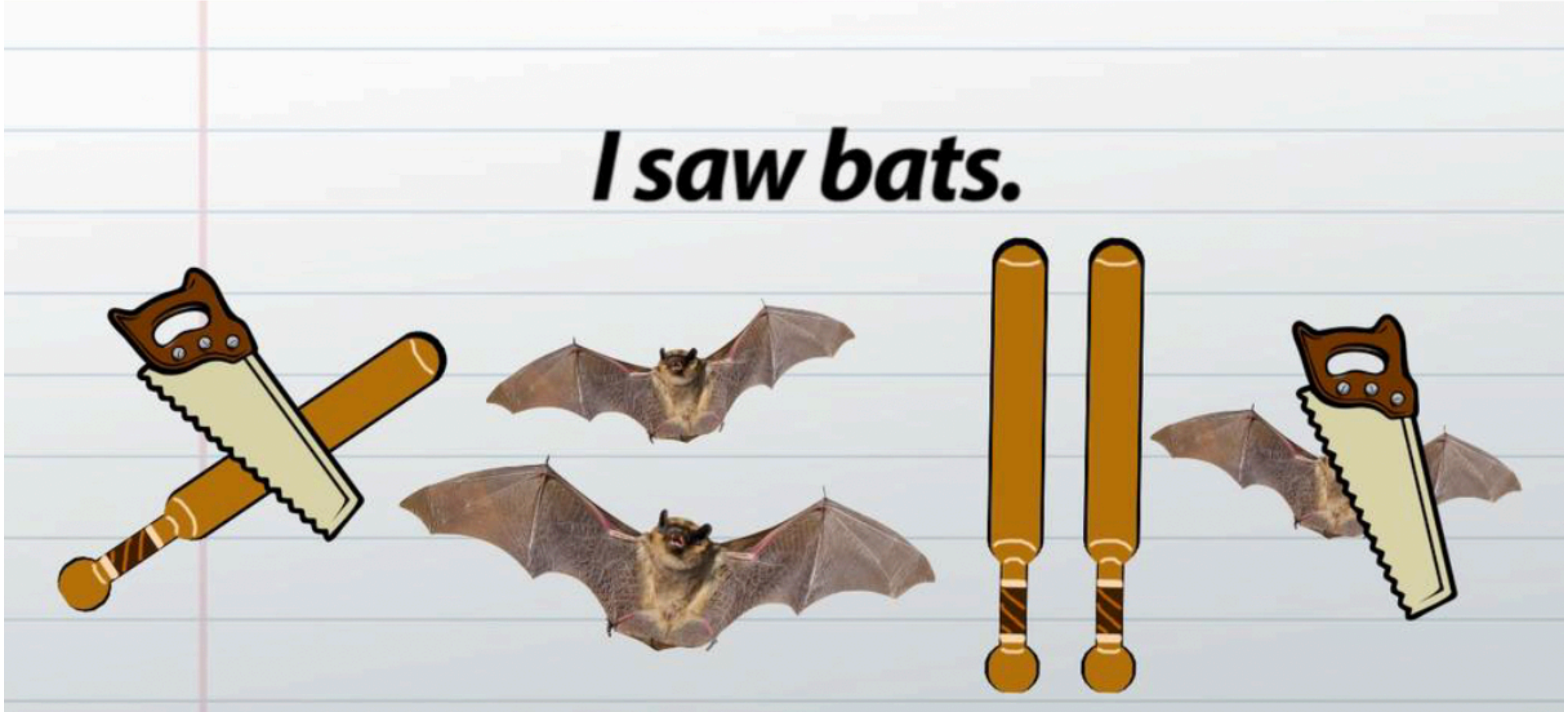
What is the meaning of a word?

Stone → Black, Brown, Hard, Hill, Mountain, Heavy, Round, Solid, ...

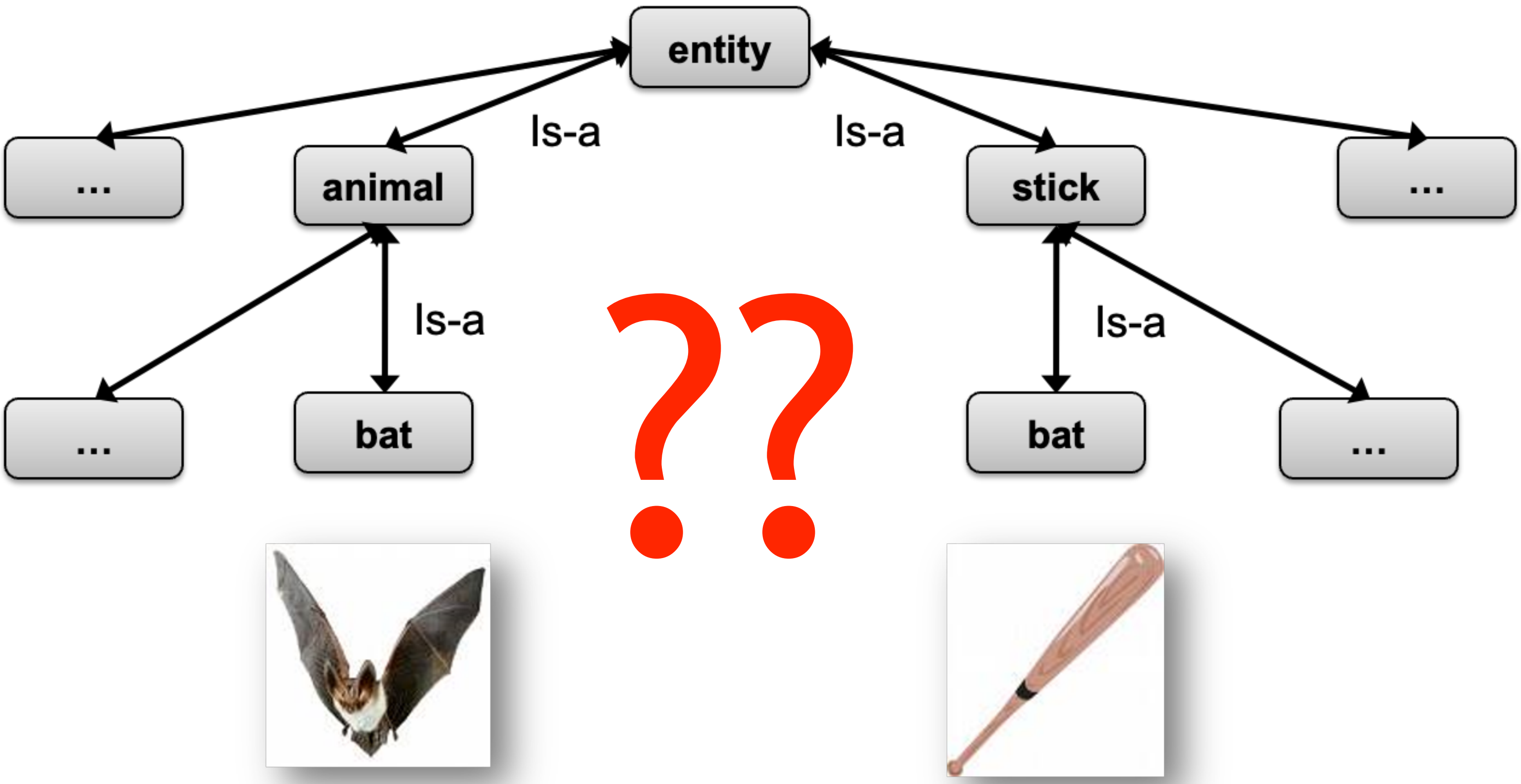
Unicorn → Animal, Unreal, Horse, White, Horn, Beautiful, ...

Greed → Money, Food, Gold, Sibling, Food, Algorithm, Thief, Politician, Police, ...

Ambiguity



How to Represent Meaning in a Computer?



Word Sense

- A **sense** (or word sense) is a discrete representation of one aspect of the meaning of a word.

mouse^1 : .... a mouse controlling a computer system in 1968.
mouse^2 : .... a quiet animal like a mouse

bank^1 : ...a bank can hold the investments in a custodial account ...
bank^2 : ...as agriculture burgeons on the east bank, the river ...

Word Sense

- From dictionaries and thesauruses
- From Running Corpus - Distributional Hypothesis

mouse^1 : .... a mouse controlling a computer system in 1968.
mouse^2 : .... a quiet animal like a mouse

bank^1 : ...a bank can hold the investments in a custodial account ...
bank^2 : ...as agriculture burgeons on the east bank, the river ...

Word Sense

- From dictionaries and thesauruses
  - ✓ give textual definitions for each sense called glosses

$bank^1$: financial institution that accepts deposits and channels the money into lending activities

$bank^2$: sloping land (especially the slope beside a body of water).

Relations Between Senses

- synonymy
- antonymy
- hypernymy
- meronymy

…….

Synonymy

- Two senses of two different words are identical, or nearly identical, we say the two senses are synonyms.
  - ✓Couch/Sofa
  - ✓Beautiful/Gorgeous
  - ✓Happy/Joyful

Synonymy

- *How big is that plane?*
- *Would I be flying on a large or small plane?*

big** has a sense that means being **older or grown up

- *Miss Nelson, for instance, became a kind of big sister to Benjamin.*
- *Miss Nelson, for instance, became a kind of large sister to Benjamin*

Antonymy

- Two senses can be antonyms if they define a binary opposition or are at opposite ends of some scale.
  - ✓long/short , big/little , fast/slow , cold/hot , dark/light
- Two senses describe change or movement in opposite directions.
  - ✓rise/fall , up/down

Food for thought

Automatically distinguishing synonyms from antonyms can be difficult.

Hypernymy/Hyponymy

- A word (or sense) is a hyponym of another word or sense if the first is more specific, denoting a subclass of the other.
  - ✓ **car** is a hyponym of **vehicle**
  - ✓ **dog** is a hyponym of **animal**,

- Conversely, **vehicle** is a hypernym of **car**, and **animal** is a hypernym of **dog**.

Meronymy/Holonymy

- a **wheel** is part of a **car**.
  - ✓ wheel is a **meronym** of car
  - ✓ car is a **holonym** of wheel.

WordNet: A Database of Lexical Relations

- Fellbaum, 1998
- English WordNet consists of three separate databases
  - ✓one each for nouns and verbs
  - ✓a third for adjectives and adverbs;
  - ✓closed class words are not included.
  - ✓ Each database contains a set of lemmas, each one annotated with a set of senses.

WordNet: A Database of Lexical Relations

- The WordNet 3.0 release has
  - ✓117,798 nouns
  - ✓11,529 verbs
  - ✓22,479 adjectives, and 4,481 adverbs
  - ✓ The average noun has 1.23 senses
  - ✓ The average verb has 2.16 senses

https://wordnet.princeton.edu/

Example

The noun "bass" has 8 senses in WordNet.

1. $bass^1$ - (the lowest part of the musical range)
2. $bass^2$, $bass\ part^1$ - (the lowest part in polyphonic music)
3. $bass^3$, $basso^1$ - (an adult male singer with the lowest voice)
4. $sea\ bass^1$, $bass^4$ - (the lean flesh of a saltwater fish of the family Serranidae)
5. $freshwater\ bass^1$, $bass^5$ - (any of various North American freshwater fish with lean flesh (especially of the genus Micropterus))
6. $bass^6$, $bass\ voice^1$, $basso^2$ - (the lowest adult male singing voice)
7. $bass^7$ - (the member with the lowest range of a family of musical instruments)
8. $bass^8$ - (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

Synset

- The set of near-synonyms for a WordNet sense is called a synset (synonym set)
- Synset represents a concept as lists of the word senses.

fool^2, gull^1, mark^9, patsy^1, fall guy^1, sucker^1, soft touch^1, mug^2

- **Gloss**: a person who is gullible and easy to take advantage of.
- Glosses are properties of a synset, so that each sense included in the synset has the same gloss and can express this concept.

26 lexicographic categories for nouns

Category	**Example**	**Category**	**Example**	**Category**	**Example**
ACT	*service*	GROUP	*place*	PLANT	*tree*
ANIMAL	*dog*	LOCATION	*area*	POSSESSION	*price*
ARTIFACT	*car*	MOTIVE	*reason*	PROCESS	*process*
ATTRIBUTE	*quality*	NATURAL EVENT	*experience*	QUANTITY	*amount*
BODY	*hair*	NATURAL OBJECT	*flower*	RELATION	*portion*
COGNITION	*way*	OTHER	*stuff*	SHAPE	*square*
COMMUNICATION	*review*	PERSON	*people*	STATE	*pain*
FEELING	*discomfort*	PHENOMENON	*result*	SUBSTANCE	*oil*
FOOD	*food*			TIME	*day*

Some Sense Relations

For Noun

Relation	Also Called	Definition	Example
Hypernym	Superordinate	From concepts to superordinates	$breakfast^1 \rightarrow meal^1$
Hyponym	Subordinate	From concepts to subtypes	$meal^1 \rightarrow lunch^1$
Instance Hypernym	Instance	From instances to their concepts	$Austen^1 \rightarrow author^1$
Instance Hyponym	Has-Instance	From concepts to their instances	$composer^1 \rightarrow Bach^1$
Part Meronym	Has-Part	From wholes to parts	$table^2 \rightarrow leg^3$
Part Holonym	Part-Of	From parts to wholes	$course^7 \rightarrow meal^1$
Antonym		Semantic opposition between lemmas	$leader^1 \Longleftrightarrow follower^1$
Derivation		Lemmas w/same morphological root	$destruction^1 \Longleftrightarrow destroy^1$

For Verb

Relation	Definition	Example
Hypernym	From events to superordinate events	$fly^9 \rightarrow travel^5$
Troponym	From events to subordinate event	$walk^1 \rightarrow stroll^1$
Entails	From verbs (events) to the verbs (events) they entail	$snore^1 \rightarrow sleep^1$
Antonym	Semantic opposition between lemmas	$increase^1 \Longleftrightarrow decrease^1$

Hypernymy Chain (Bass)

```
bass³, basso (an adult male singer with the lowest voice)
=> singer, vocalist, vocalizer, vocaliser
   => musician, instrumentalist, player
      => performer, performing artist
         => entertainer
            => person, individual, someone...
               => organism, being
                  => living thing, animate thing,
                     => whole, unit
                        => object, physical object
                           => physical entity
                              => entity
bass⁷ (member with the lowest range of a family of instruments)
=> musical instrument, instrument
   => device
      => instrumentality, instrumentation
         => artifact, artefact
            => whole, unit
               => object, physical object
                  => physical entity
                     => entity
```

Graph view of Wordnet